

[00:00] Hello, everyone.

[00:34] How are you doing today?

[00:37] Good.

[00:37] How about you?

[00:39] Very, very good.

[00:40] Lachlan, how was the day?

[00:41] How was the weekend?

[00:44] Uh, weekend was pretty good.

[00:47] Actually, last week was kind of eventful for me.

[00:50] I got a- Oh, yeah.

[00:52] I've been in the process of applying for a role with the Defense Force for after I graduate since about January last year when I was looking for my internship.

[01:03] Mm-hmm.

[01:04] And I got my proper job offer last week.

[01:07] Congratulations.

[01:08] This is...

[01:09] This should be party time then .

[01:10] Yeah.

[01:11] Well, not party time.

[01:12] I don't actually.

[01:14] I'm, I'm st- All right ...

[01:14] I'm not enlisting until March, but so it's still- Yeah ...

[01:17] so it's still months away, but, you know.

[01:19] All right.

[01:21] Uh, let, let us know a bit more about the process and everything.

[01:24] We, we all want to hear from each other.

[01:27] Go ahead.

[01:28] Um, so the role I'm enlisting for is a networks operator with the Navy, so I'll be, uh, responsible for, um, well, making sure, like, the IT networks on the ships between the bases and stuff, um, works and facilitating communication between

[01:50] them.

[01:51] Um, the role itself has, I think about a six-month, a little over, like six to seven-month training period, um, and then two years of minimum service.

[02:04] Oh, that's excellent.

[02:06] That's very excellent.

[02:07] Mm-hmm.

[02:09] So what, what do you think, like, a, like a w- what, what skills exactly work for you in this role?

[02:24] What, what, what actually helped you to stand out?

[02:27] Or what do you think which, "This is the, this is the qualification or this is the skill that I had, and that's why they have chosen me."

[02:34] Is there anything specifically?

[02:37] Um.

[02:37] Yeah.

[02:39] No, I, I'm not entirely sure.

[02:43] It might be because of my history with, um, Scouting Australia, 'cause I was a part of that for several years.

[02:51] But it could also, um, it could be, uh, my specific IT skills, though, again, I'm not entirely sure.

[03:02] Uh- W- what, what was the thing you said, the history with...

[03:06] What Australia group?

[03:06] Uh, Scout- Scouts Australia.

[03:08] Oh, oh.

[03:09] I was a part of the Australian Scouting Association- Makes sense.

[03:10] Yeah, yeah, yeah ...

[03:11] for several years.

[03:11] Scouts As- Okay.

[03:12] Oh, all right.

[03:13] All right.

[03:14] Okay.

[03:14] That could be one thing.

[03:15] Yeah.

[03:16] Not bad.

[03:17] So, hmm, that's good.

[03:19] That's good.

[03:19] So, yeah, so how was the process?

[03:23] Like, how many times or what, or how many types or how many times of interview they have taken from you?

[03:29] What, what are the kinds of interviews?

[03:32] So the process, so it, as I say, started back in, um, January, where I went through and the initial, um, application process was me submitting, basically, like I submitted my resume, a cover letter, and I went through and gave them,

[03:51] like, my general history.

[03:52] So I told them about my history in Scouts, what I w- how far along I was in my course at the time, that kind of thing.

[04:00] Mm-hmm.

[04:00] Um, then after that, I, once you get, uh, accepted as a candidate- Mm-hmm ...

[04:05] you have to submit an, you have to keep submitting an up-to-date version of your resume.

[04:11] So if, like my, I'd started my internship and completed it over the course, so I had to update my resume with that.

[04:18] Um, I then have to submit, like, various other documentation, like birth certificate and all this kind of stuff, official stuff to prove who I am.

[04:26] Mm-hmm.

[04:27] Then, um, once you've completed, once that, um, is part, you, during that period you also choose, um, like what roles you wanna go with.

[04:40] At the...

[04:41] You're also are assigned, uh, or put in contact with a, um, person in the recruiting section of the Defense Force, and they will help you.

[04:50] You'll be in regular contact with them, discuss, like, what roles you wanna, you wanna aim for, your preferences, that kind of thing.

[04:58] Um, once you've reached the interview stage, there's three types of interviews, um, that I had, that I had to go through.

[05:10] Um, they were, um, a medical one, so I have to make sure that I'm in proper- Gotcha ...

[05:16] medical health.

[05:17] Mm-hmm.

[05:17] There's a s- there's a psychological interview- Hmm ...

[05:20] um, where they have you sit down with a psychologist and go through, um, and the psychologist talks you, um, through things, make sure you understand, make sure you're of sound mind- Hmm ...

[05:33] that you are not, that you are making this choice, that you kind of understand the import, that you are, um, that you don't have, like, any, potentially any underlying conditions that could, um, make you, uh, I don't wanna say a

[05:48] threat, but like ineffective in the r- in the, in the Defense Force.

[05:53] Hmm.

[05:53] And then there's, um, the actual, they call it the defense interview, but it's the job interview.

[06:00] You will complete, um- An interview for each, in, in the same time, um, in a row, you'll, for each of your preferences.

[06:12] Um, and so- Mm-hmm ...

[06:12] they'll go through, they'll ask you about, um, the role, they'll talk a bit, you'll talk about the stuff.

[06:19] Um, then depending on branch and high, how high up the position, there's potentially two other, um, kind of interviews if...

[06:31] So my role with the Navy does not have, I do not have to pass a physical assessment beforehand.

[06:38] Mm-hmm.

[06:38] But, um, there, but most other roles do, the Navy as- Yes ...

[06:43] the, in the Army and the Air Force.

[06:46] Right.

[06:46] Um, I have, uh, I participated in one of those, so I know I would pass it, but I ha- but I do not have to for my- Mm.

[06:55] ..

[06:55] role specifically.

[06:56] Mm-hmm.

[06:56] And then if you are going for an officer role, you have to, um, complete basically a screening where you will sit down in front of a panel of officers and assessors, and they will assess whether or not you are suited

[07:13] for that, for the officer, um, role.

[07:16] It's far more rigorous and I wa- initially I had an officer role on my preferences.

[07:21] Mm-hmm.

[07:22] I was told, "We don't think you're ready for it.

[07:25] We recommend you- Mm ...

[07:27] st- uh, you choose a different role for your primary focus, um, and potentially you can get promoted later."

[07:35] Um- Mm.

[07:35] Mm ...

[07:36] and this f- uh, after you've completed, um, your, those assessments, um, the results for those, uh, so the medical results last two years, and the Defense Force, and the job interview and medical and psychological interview results last one year.

[07:56] Um, so I completed my initial round of all of those in August last year.

[08:02] Mm-hmm.

[08:02] And I, I, in that time period, after you've completed your interviews, you're assigned an enlistment coordinator, 'cause you're- Mm-hmm ...

[08:11] getting to the tail end at that point.

[08:13] And the enlistment coordinator will be assigned based on which branch you are going to enter.

[08:19] So I have a, so my enlistment coordinator is part of the Navy because that's the, uh, branch I'm going into.

[08:27] Mm-hmm.

[08:27] And, uh, they will help you organize, uh, getting your security clearance update.

[08:34] You have to go through the- Right ...

[08:36] Australian Government Security and Vetting Org- Right ...

[08:39] Agency.

[08:40] I think that is NV1 or something like that?

[08:43] Yes.

[08:44] Yeah.

[08:45] Okay.

[08:45] Uh, N- I think I'm NV2 is my security clearance now.

[08:49] Okay.

[08:49] Okay.

[08:49] Um, and so you'll have to go through and get your sec- security clearance updated and then submit that to the Defense Force so they know you've, that you've been properly vetted.

[08:59] Um, and then after that's done, you've got, uh, because I'm completing my course before I enlist, I have to, um, I had to redo my defense and psychological interview because the, uh, results would, do not, uh, last.

[09:20] So you have to redo those, yeah, ev- when the results expire.

[09:25] Um, and then after you've got, they've got all your information, they've got all your results, you can lock in a enlistment date.

[09:34] Um, for me, that's 15th of March next year.

[09:37] Mm-hmm.

[09:37] And they will, um, then send you the official job offer, where you have to- Wow ...

[09:44] uh, go through...

[09:45] There's two forms I had to submit, which were, uh, you have to, a signed job let- uh, or letter of offer.

[09:54] Mm-hmm.

[09:55] Um, you have to s- you have to sign that, and you have to have a witness sign it as well.

[10:00] Mm.

[10:01] Um, if you are enlisting as an underage person, you have to have both your parents sign it.

[10:07] Mm-hmm.

[10:08] Um, since I'm not underage, it was just me and one witness.

[10:13] And then you have to give them, um, like all the necessary details, like your address, your, like your current address, your next of kin, your emergency contacts, your tax file number, your bank account information, all that kind of stuff so

[10:27] they can, you know, pay you and stuff- Oh, yeah.

[10:29] Yeah, yeah ...

[10:29] once you actually get in.

[10:31] Um, and yeah- Mm-hmm.

[10:32] So- Cool ...

[10:33] the next stage for me is I've got to prepare for my enlistment in March.

[10:38] Mm-hmm.

[10:39] Um, at which point when, on my, the day of my enlistment, I will be given a final, um, uh, like contract to sign that, that will properly enter me into the Defense Force.

[10:53] All right.

[10:54] Cool.

[10:55] I think, Lachlan, that's, that's a good, good experience, I'd say.

[10:58] Um, a question, actually, I'm asking all these questions to your, letting you to describe whole s- whole story in this session, and eventually it is recorded, which is great for other students too, actually, who are looking for this kind of

[11:12] opportunities, and that's why I, I'm asking you and giving these things.

[11:17] Um, I think sometimes we, we need to really know and share this, our thoughts or our way or our process and our journey to others so that others can benefit out of it.

[11:31] So this is fantastic actually.

[11:33] Um, I'm just wondering, Lachlan, actually, I'm trying to understand the whole thing.

[11:38] So you, if, if I am correct, you are already in the bachelor, I mean bachelor of IT or something, some of your major here at Deakin, right?

[11:52] Yes.

[11:53] How does it work with the, your, your job enrollment?

[11:56] Uh, you're not going to finish your graduation, right?

[12:00] Uh, so, no, thi- this is because of my circumstances.

[12:05] This is my last unit.

[12:07] I've completed every other...

[12:09] So this is- Oh ...

[12:11] this is an, this is what, because of something that happened to me.

[12:14] I completed my...

[12:16] When I graduated in 2020, I took a six-month break, then started a diploma of IT at Deakin College.

[12:24] And then from there I moved into my current bachelor.

[12:28] Um, and then in 2024 as I was getting towards, I was hitting like the la- later half of my bachelor, I ran into an issue with my, um, course where I'd taken too many level one elective, uh, units, and so

[12:49] I had to, uh, talk with Deakin, with Student Central and get like my prior learning and stuff reorganized so that I had enough, 'cause I would have gone over the amount of credits needed.

[13:01] Mm.

[13:02] Um, and so I was supplied with a new, um, unit plan with, uh, all the, and it was like, "These are the units you now have to complete at this, and this is the time we recommend."

[13:16] But the way it was formatted, they had this unit up with all the units I'd already completed, which were- Yes ...

[13:22] all my level one units.

[13:23] Got it.

[13:23] So I didn't see it until, uh, earlier this year.

[13:29] Mm.

[13:29] I completed my last level three unit last trimester.

[13:33] Mm.

[13:34] I got it.

[13:35] I got it.

[13:35] And so I wai- And so it got to the time when I should have received my graduation invitation- Right ...

[13:41] and I didn't get it.

[13:42] And I'm like, "Why?"

[13:43] I get home, check Student Central, and I'm only at 23 out of 24 units.

[13:48] And I'm like, "Oh, that's annoying."

[13:50] So I'm- Mm-hmm.

[13:53] Makes sense ...

[13:53] so yeah, this is my last, this is my last year.

[13:55] Makes sense.

[13:56] Makes sense now.

[13:57] I just ...

[13:58] Yeah, no.

[13:58] That's a funny thing.

[13:59] But they- All right.

[14:00] I got it ...

[14:01] they, yeah, as from what I understand with the, um, and the people I talked with when I went for my, with my defense interview- Yep ...

[14:10] uh, or my initial set of interviews with the Defense Force, uh, back in August last year, most of the, they're pretty long processes.

[14:18] You and everyone I talked to, all of the, um, members of the force who I've been in contact with bec- as, uh, like liaisons, they've told me, "Finish your course first so that you're not, um, you're not like you get

[14:35] partway through, go off, spend for me what would be two to three years, um, as a service member, and then come back and have to pick up my course wherever I left off."

[14:47] They say, "Finish it first and then enlist."

[14:50] Mm.

[14:51] Yeah.

[14:51] Yeah.

[14:52] I think, I think, I think that's a good suggestions, and, and that, that purely makes sense.

[14:56] And I think process of defense is of course that the whole process, interviews, and all the stages has to be really, uh, how can I say?

[15:07] Very thorough, and also it has to be really in a, in a way, uh, it's they are recruiting someone for the defense, so defense of Australia, so need to be really, uh, went through the proper way.

[15:20] And, and I think that's a good, good things.

[15:23] I think, I think eventually the pay should be good and everything, all the other facilities should be very good, um, I believe.

[15:32] Um, uh, so, so eventually from your role when you'll be promoted and, and slowly, gradually when you will be, uh, I mean, going up in the ladder, so where, where you will reach or where you see like from today or

[15:48] from now to 10 years' time?

[15:50] How, how it will be?

[15:54] Um- Any, any trajectory was shared with you or do you have any?

[15:57] I've not, I haven't, this isn't something I've discussed with my, uh, enlistment coordinator or any- Yeah ...

[16:05] of my liaisons currently.

[16:07] The reason I initially went into the Defense Force is because I do, uh, I wanna work in IT- Mm ...

[16:14] but the current IT job market is really bad because of the disruption that AI is causing- Mm ...

[16:21] especially at, at the junior level.

[16:24] Yeah, yeah.

[16:24] That's right.

[16:25] You've- I'm sure certain you've heard the stories of companies like freezing their hi- hiring of junior level developers for- Right, right ...

[16:32] and you- Right ...

[16:33] replacing them with AI.

[16:34] Right.

[16:34] So it, it's ma- it's obviously meant that it's incredibly hard for if I was just a new graduate to find- Mm ...

[16:44] a, um, a entry, a job f- for my level within- Mm ...

[16:48] the industry I wanna work in.

[16:49] Mm-hmm.

[16:49] Mm-hmm.

[16:50] And so the Defense Force is, um, was a role that I had already kind of been eyeing because when I was looking for an internship for- Mm-hmm ...

[17:01] that, that had come up.

[17:02] The Defense Force had an internship program- Mm.

[17:05] Mm-hmm ...

[17:05] that I had applied for.

[17:06] I hadn't gotten in, but- Mm ...

[17:08] I, they put me on like a wait list.

[17:10] Mm-hmm.

[17:11] And so I went, as I was reading through it, I realized, and I saw the job opening.

[17:17] Mm.

[17:17] I realized this is something that gets me job experience.

[17:21] Lots, it looks really g- it can look really good on my resume.

[17:25] Mm-hmm.

[17:25] And it is a stable role I can have for f- in, for this role two years.

[17:32] And if I don't, if I no longer want to work the role, I can leave the Defense Force, go into the reserves, and get a civilian job.

[17:41] Yeah, yeah, yeah.

[17:42] However, if I don't, then I have a guaranteed path of progression.

[17:47] It gives- Correct ...

[17:48] it, so it gives, it, I went into it not, both because it is in an industry that I wanted to work in and because it has a stability the current job market- That's right ...

[17:57] doesn't- Yes ...

[17:58] can't provide me.

[17:59] That's right.

[18:00] That's right.

[18:01] That's right.

[18:01] I think, I think, I think that's a, that's a good perspective and good way to, good way to decide and navigate.

[18:08] But one thing, yeah, I, I just want to maybe this session is actually, uh, there is another, another reason I'm- Probably discussing with you all these sort of things.

[18:18] Today we have, we have only HD, HD task discussion.

[18:24] There is nothing else.

[18:26] And, uh, HD task, actually, I basically I'm not going to code anything.

[18:30] I'll, uh, I'll discuss with you.

[18:32] And I believe those who are actually claiming for HD, I don't need to discuss much with them because they already claimed themselves for HD, so they should be able to do better.

[18:41] Uh, but anyway, let's continue because, uh, that is the only opportunity we get to converse, and we got to do some conversations.

[18:49] So actually Lachlan, your, your, your, your...

[18:53] the way you are deciding on from your point is really, really good.

[18:56] But one thing I want to highlight here is all about the AI disruptions.

[19:03] That's absolutely true.

[19:04] There is no doubt about it.

[19:05] AI is de- disrupting all the, all the, um, industries.

[19:12] Um, and a lot of recruitment are freeze at the moment.

[19:18] Also, there are a lot of job cuts we can see.

[19:22] But let me tell you one very important thing.

[19:25] Uh, these jobs will eventually be changed or modified in a way people will be es- using their skills in another way, not just typing or not just coding stuffs.

[19:41] But IT will be there, and AI also will be there.

[19:46] And we m- we also need to know the fundamentals of IT and AI.

[19:54] That's important.

[19:56] And eventually I would -- I see myself, like, in, in next couple of years, um, the, the roles will be changed.

[20:05] Also, the skills will be changed, but it is -- it, it will be needed.

[20:10] And eventually AI will enter to many aspect of our life in daily research, daily activities, and all sorts of things.

[20:20] So IT, we need to know the fundamentals of IT or how underlying things work.

[20:25] Currently at the -- at this point, some people are very much affected are the one who are actually working as a software developers at different companies.

[20:34] The biggest -- the big reason is last twenty, twenty-five years of time, we had a huge influx of developers because companies needed them.

[20:43] So software developers or s- or developers or programmers, we needed a lot to develop those sort of things.

[20:50] Since AI came, AI can, uh, can quickly do this job, that's why we don't need too many of these developers or, or programmers.

[20:58] But, uh, eventually people having those, again, unique skills, they will be remaining, but those who are doing pretty much similar things, they will be replaced by AI.

[21:08] However, there will be n- another way of, another way of using IT will be -- people will explore, and eventually that will go flood all the other areas.

[21:20] This is one of the key thing I would say.

[21:22] That's why when this kind of disruption happens, normally s- little bit, uh, I mean, things get freeze and also some, some s- s- certain group of people are normally affected.

[21:35] Here in this case, I would say the developers are affected.

[21:39] But one -- another thing I would like to highlight here, because many students who are studying IT, they are getting a bit worried, and I, I, I see this question very much.

[21:48] Actually, last, last week or week before, I attended Beyond the Degree panel discussion at Deakin.

[21:56] So Deakin Student Association has invited me to attend and become a panel member of that, um, uh, that Beyond the Degree session.

[22:08] And, and this is one of the things I want to highlight is when -- actually imagine when calculator has been invented.

[22:17] So people can do all the calculations.

[22:21] People can, uh, do all this, um, two plus two is four, two plus five is seven.

[22:27] We do the calculation, but why at a school we still teach kids two plus two?

[22:35] We don't give them calculator to do that job.

[22:38] Eventually, if I give a kid to do and show him, "Okay, this is two, this is plus sign, and this is, uh, five, so find the result," they can immediately do it.

[22:48] And you have done Naplan test, you've done a lot of test.

[22:52] Uh, so we are, we are, we are learning those fundamental at our primary schools, right?

[22:58] And we are not leaving them.

[22:59] So those fundamentals we eventually need to know because whatever AI we call it, even if you want to use it, use those AI in a secure way, in a proper way, in a jus- in a proper judgmental way, like I

[23:13] need to judge what the AI is giving me, I need to understand those fundamentals.

[23:18] This is one of the key part of things.

[23:20] And majority of the, of the AI -- of, of the, um, um, the, the video clips that we see in YouTube or in Facebook or eem- even in different channels, those -- the top-level people when they talk about AI and

[23:42] they said that, "Hey," like, like if you, if you see that talk from Sundar Pichai or if you see some, uh, the, the CTO of Anthropic, if you see they are saying, "Oh, our AI -- our, all our codes are

[23:55] actually written by AI."

[23:57] Last, last few months, ninety percent of the, of the codes are actually, um, ninety percent of our codes are actually, uh, uh, our codes are actually developed by AI.

[24:10] We haven't done anything.

[24:11] Look, let me tell you one thing.

[24:13] When they are saying our codes are developed by AI, but those people who are telling you that story, eventually those -- they learned Developing or they, they learn coding by just pen and paper, by doing the raw coding and fundamentals

[24:30] of coding.

[24:31] So people who are giving those hype or things to the other people, eventually they themselves actually, uh, wrote those code and they understand the code very well.

[24:41] That's why they can say that, "Oh, okay, that code is actually written by AI."

[24:45] But we know when we are saying, "Okay, AI came here, and then we are using AI."

[24:51] But you as a, as a, as a developer or as a programmer or if I want to build something, I need to know some fundamentals.

[24:59] Without those fundamentals, I will not be able to build.

[25:02] Even AI giving me code, I will not be able to actually customize it or do something.

[25:07] Okay?

[25:08] So this is the biggest thing actually people are not...

[25:12] true message is not passing, uh, to the, to the Facebook channels or other places.

[25:18] And, and that's why I, I can see for few years that things will be frozen, some people will be disrupted, but eventually those job will be again created in another form.

[25:30] That's what I can clearly see, uh, from my, uh, point.

[25:35] And also, and that's what I strongly believe.

[25:38] Because if that is the case, there is no point in our primary school or in our high school we are teaching you algebra, we are teaching you fundamentals of mathematics.

[25:48] See, we have already calculator is there.

[25:50] We have everything is there.

[25:52] Yeah?

[25:53] And top universities in the world, you will be surprised to know that top universities in the world, they are even, even taking the test with hard copy, with the pen and paper.

[26:06] I'll give you one example, just to few of you are here, but always we have this question.

[26:12] So I'll give you one example.

[26:14] I had an opportunity like, uh, just couple of weeks back, I had an opportunity take an in- take a jo- take a, a, a test or invisi- invigilator test from Zurich University, Switzerland.

[26:26] So as you know, at Switzerland, the Zurich is the number one university in the, i- in that country.

[26:33] And it is one of the top-ranked university.

[26:35] So one of the student from Australia, he was a student of Zurich, and he had to somehow move to, uh, move back to Australia here.

[26:44] So I, m-, uh...

[26:47] So, so they need to actually conduct the test.

[26:50] So they could actually conduct the test online.

[26:52] There is no, no issues at all.

[26:53] But they have, they have actually arranged the test in a way that I will be the invigilator for that student.

[26:59] I had to book a room.

[27:01] I...

[27:02] They, they send me the question paper just right before the exam, not even few weeks back, just right before the exam.

[27:08] And the job that I had to do is literally print that question paper, give...

[27:13] And that was a bachelor student.

[27:15] He was actually, actually a student of IT.

[27:18] Yeah.

[27:18] I, um, I...

[27:21] He was a student of IT and, and mathematics or something like that.

[27:25] So it's a combination of two.

[27:28] So his degree was something like that.

[27:30] And he was a student of actually Monash University.

[27:32] So the test was happening in a way that I had to print and give the student that question.

[27:38] The clock was ticking, okay, 9:00 to 11:00.

[27:40] This is a two hours time.

[27:42] And I'll be...

[27:42] I, I, I was sitting and the student was giving the test in front of me.

[27:46] I was literally giving him a physical invigilation.

[27:50] And right after the, the, the question was printed and he was actually answering in pen and paper, not in the system in computer.

[27:59] So I had to literally scan that res- response and send back to Zurich.

[28:04] Also, I had to post that, that, uh, response, uh, from Australian post to send back to the professor at, at Zurich University.

[28:13] So that's how those universities are considering.

[28:16] Why they are considering it?

[28:17] The reason behind it, they know that we have to teach our kids fundamentals.

[28:22] Without fundamentals, whatever AI magic we see, that's not gonna work.

[28:26] And eventually, uh, if we want to protect all of us in Australia as a territory or our data, our, uh, our coding or our programming or our software, we have to learn those things.

[28:39] Without that actually, AI not gonna help us.

[28:41] Rather it will create a lot of security threats.

[28:44] So that's my observation.

[28:46] And I would say, uh, that's gonna happen eventually.

[28:50] So, so don't worry too much.

[28:52] Um, even if you are studying IT, if you go underlying principles of IT, that will remain forever.

[28:58] That's the key part of it.

[29:00] And the other thing I want to say here, since you are in artificial intelligence and, uh, and machine or data science and artificial intelligence or artificial intelligence or machine learning, so eventually the fundamentals of these are actually some, some formulations

[29:19] of math you will see, um, or, or visualizations or data cleaning or, or plotting.

[29:27] So they all require actually, uh, a lot of, uh, work, which AI probably can...

[29:33] You can use AI as an aided tool, but actually the fundamentals of those or underlying principles of those we need to know.

[29:42] So again, people who will be focusing on that is extremely important.

[29:47] Like, um, they will, they will, um, they will be there.

[29:53] Yeah.

[29:54] So yeah, I think that's, that's it.

[29:57] Yeah.

[29:57] A bit of, a bit of, uh, off-topic discussions.

[30:00] Yeah.

[30:01] Anyway, um, Catherine, how have you been and how is it going with you?

[30:07] Catherine?

[30:09] Yep.

[30:09] Uh, yeah, not too bad, thank you.

[30:11] Not as exciting as Lachlan's week.

[30:14] Um- It will be coming up for each of us at different weeks.

[30:17] Anyway, go ahead.

[30:19] Uh, I mean, nothing too exciting.

[30:21] Uh, I was trying to smash out some- Assignment stuff over the weekend- Oh, wow ...

[30:25] especially when the weather was pretty bad.

[30:27] Right.

[30:27] I live in Geelong, so weather was pretty bad down here, so trying to get a little bit ahead on that.

[30:33] Better.

[30:33] Better.

[30:34] I think better if you can, uh, when weather is bad, you try to complete all your things.

[30:39] When weather is good, you go outside and rock and roll.

[30:41] That's better for all of us.

[30:43] That's the plan.

[30:44] Yeah, yeah.

[30:46] Sophia Rolling, how have you been?

[30:47] How was your weekend?

[30:50] I've been pretty good, thanks.

[30:52] How are you?

[30:53] I was, I was doing great.

[30:55] It was fantastic- Yeah ...

[30:56] for me.

[30:57] Anything, any exciting news?

[30:58] Anything you are doing?

[31:01] Um, not really.

[31:03] Okay.

[31:04] Just trying to manage and doing, do...

[31:06] catch up with all the assignments, right?

[31:08] Yeah.

[31:09] Okay.

[31:10] Okay.

[31:10] All right.

[31:11] Tonky, how are you doing?

[31:14] How have you been?

[31:15] I know you are working hard for HD, I guess.

[31:18] Am I right?

[31:19] Oh, no, no.

[31:20] I'm, I'm just- ...

[31:21] coming for credit.

[31:22] I'm doing well.

[31:23] How are you?

[31:24] Oh, okay.

[31:24] Good, good.

[31:25] I'm doing great so far.

[31:27] Um, so what...

[31:28] how was your weekend?

[31:29] How was the things?

[31:30] Um, not too bad.

[31:31] Just catching up, doing a lot of assessments and- I know ...

[31:35] just trying to catch up right now.

[31:36] I know.

[31:36] Hopefully I'll, um, I'll finish on time, so that's about it.

[31:41] Right.

[31:42] Right.

[31:43] Right.

[31:43] I think, I think one of the key thing we learn through this process is during this week, we are very much busy with our assignments and deadlines.

[31:50] So this is one of the things we learn in the univ- university.

[31:54] Basically, when you go to industry, even in any, any workplace, even if you do your...

[32:00] if you, if you are your own entrepreneur, if you are becoming a...

[32:02] trying to become an entrepreneur, so eventually you will have things.

[32:07] Sometimes it will be very flexible, and sometimes it will be too tight.

[32:11] So how you manage your time?

[32:13] I always suggest everyone, during this time of the week, we are super busy, so we all need to be, like, careful and manage our time well.

[32:22] That's the only thing we can do.

[32:24] So, and that's how we learn.

[32:27] Paulo, how are you doing, Gabriel?

[32:31] How was your weekend?

[32:32] I have been, been great.

[32:34] Tell us more.

[32:36] Uh, I just spent my whole time for doing assignment.

[32:42] That's it?

[32:43] Yeah, because- No other, yeah ...

[32:45] I'm still adjusting myself with Australian education system because- Ah, okay ...

[32:50] you know, uh, my country, the education is not advanced like Australia, so yeah.

[32:55] Absolutely.

[32:56] Absolutely.

[32:57] This is, this is fully understandable.

[32:58] Paulo, y- so you are first trimester or second trimester?

[33:03] Which trimester for you, this one?

[33:05] Uh, now is trimester two, but- Ah, okay ...

[33:09] this is my first year.

[33:11] So this is pretty much first year.

[33:12] This is, this is fully understandable.

[33:14] We, we...

[33:15] that's why sometime I always say when I, uh, when I try to deliver my lecture, I will be eventually delivering lecture for this unit next trimester, T3.

[33:25] So T3 I'll be lecturer for this unit, actually.

[33:29] So what I have seen, I...

[33:32] we always keep in mind that, okay, even I'm not, I'm not local, right?

[33:36] I, I, I just, I just migrated in Australia, so you can see I'm not a native person, so we need to cope up with a lot of things.

[33:43] We need to, first of all, cultural barrier, language barriers, a lot of things we need to adjust, right?

[33:50] Mm-hmm.

[33:50] And I think most of us, even, even many of them migrated from Europe, many of them migrated from South America.

[33:56] I see, um, and this is part of our life, and sometimes Asia and different parts of the world.

[34:02] But adjusting, managing, learning, that's how we are human, actually.

[34:06] We...

[34:07] I- if a robot born in Us- America, if you bring it in Australia, it's same.

[34:12] There is no change.

[34:14] And if you bring that robot to China, that will be remaining same.

[34:17] But as a human, we adapt.

[34:20] We have the great level of adaptability.

[34:23] We can adapt a, a lot, and that's how human actually evolved over time and they...

[34:29] because of their cerebral capabilities.

[34:32] And, and that's how human are so intelligent, so powerful.

[34:35] Um, we can be in the cave, and we are now sending human to space.

[34:42] We are same human, right?

[34:44] So excellent, Paulo.

[34:47] Thank you.

[34:47] Which country you are from, by the way?

[34:48] Is it, is it somewhere in, um, Philippines?

[34:52] No.

[34:52] No, from Timor-Leste.

[34:55] Sorry, can you repeat again?

[34:57] Timor-Leste.

[34:57] Timor-Leste.

[34:59] Where is that?

[34:59] I never heard of it.

[35:00] Uh, it's, it's close to Darwin, actually.

[35:03] Hey, man.

[35:03] Located between Indonesia and Australia, yeah.

[35:06] What, what is that name?

[35:08] Timor-Leste.

[35:09] Ti- Timor-Leste.

[35:10] Yeah.

[35:10] Ah, I think I saw that name, yeah.

[35:12] Timor Les- hold on a second.

[35:14] Awesome.

[35:16] Can you give me a bit of your, your...

[35:17] is it an island?

[35:20] Uh, it, it was part of, um, Indonesia before.

[35:24] Mm-hmm.

[35:25] Uh, in 1975, but we already got independence since that.

[35:32] But, um- So you got, you got i- hold on a second.

[35:33] You got independence from Indonesia?

[35:36] No, no.

[35:36] Um, actually, um, we, we were, were colonized by Portuguese.

[35:42] Um- That's what I can see from your name.

[35:44] I was going to ask you, actually.

[35:45] Yeah.

[35:47] Oh, okay.

[35:47] That's why most of, uh, Timorese name we adopted from, uh, Portuguese name.

[35:51] Yeah.

[35:52] Oh, okay.

[35:53] Yeah.

[35:53] After that, uh, couple of years, Indonesia invade again.

[35:57] Yeah.

[35:58] Oh, okay.

[35:58] And now we become a country.

[36:01] It's a small country, yeah, actually.

[36:03] Okay.

[36:04] Posted a map in the chat if you want.

[36:06] Yeah.

[36:06] Yeah, I saw that.

[36:08] I saw sometime actually previously, but I forgot completely.

[36:11] Anyway, Paulo, tell me, like, a bit more, like, the, the people, the ethnicity.

[36:21] Is it ethnicity mix of, of Indonesia and Portuguese, or it's actually mostly Indonesian- Ethnicity.

[36:28] People I think it makes a lot Chinese, because I have a Chinese blood as well.

[36:34] Okay.

[36:35] Wow.

[36:35] And, yeah, some of, of them are Portuguese, some of them, uh, Australia, Japan, because, uh- Wow ...

[36:43] yeah, we were inva- invaded by many countries.

[36:48] That must be- And also- ...

[36:49] beautiful country then ...

[36:52] um, I think, uh, previously there is also, um, like, commercial activity as well.

[36:57] That's why, uh, some of, uh, Chinese ethnicity in there.

[37:02] And some of them also, uh, brought by the Portuguese, uh- Okay ...

[37:05] during the colonization.

[37:07] Yeah.

[37:07] For, uh, they, they say it's from Macau, but I'm not sure.

[37:12] Okay, okay.

[37:13] Interesting.

[37:15] Wow.

[37:16] Wow.

[37:17] So the language, what language do you speak?

[37:20] Uh, we have two, uh, official language, uh, Tetum and Portuguese.

[37:25] Really?

[37:26] Yeah.

[37:27] So, I mean, people is- still speak Portuguese over there?

[37:31] Uh, actually at the time, I think, uh, only half of people are able to speak it.

[37:38] Okay.

[37:39] Portuguese.

[37:39] Yeah.

[37:39] So, so now, now you have, you said the two official language.

[37:43] One is the Portuguese, another is the?

[37:45] Uh, Tetum.

[37:45] Uh, it's like our national language.

[37:47] Yeah.

[37:48] Oh, okay.

[37:50] Okay.

[37:50] Wow.

[37:51] Wow.

[37:51] And I believe the, the weather is a bit tropical weather, isn't it?

[37:56] Yeah.

[37:56] Yes.

[37:57] Okay.

[37:58] We, we only have, uh, two season in my country.

[38:01] Uh- What are the two seasons?

[38:02] Dry and wet season.

[38:04] Only that.

[38:04] Really?

[38:05] Yeah.

[38:05] You don't have winter?

[38:07] No.

[38:08] Funny.

[38:08] Oh, my God.

[38:10] Okay.

[38:11] O- okay.

[38:12] Now, when you travel to your, your, your, your city or your country, so you, you can take direct flight to, uh, to your city or to your country?

[38:23] Uh- How does it work?

[38:24] Or you go to Darwin and then- Yeah, to Darwin ...

[38:26] go some other way?

[38:28] To Darwin or Bali.

[38:30] Then, so you go to Bali?

[38:32] Bali- And then- ...

[38:33] and then flight, flight again to my country, Timor.

[38:36] Oh, okay.

[38:37] Or- How long it takes?

[38:38] Is it too much time it takes, or?

[38:41] I think both take the same time- Okay ...

[38:44] because, uh, uh, no.

[38:46] From Bali it's too long because, uh, six hours from here to Bali and then another two hours to my country.

[38:53] But- Oh, okay.

[38:54] But from, I think Darwin it's easier, right?

[38:56] Yeah.

[38:57] Darwin is easier.

[38:58] Okay.

[38:59] That's good.

[38:59] Yeah.

[38:59] From here four hours.

[39:01] Yeah, yeah.

[39:02] And then- From, from here, Darwin itself- ...

[39:03] one and half hour already.

[39:04] Yeah.

[39:05] Exactly.

[39:05] You are inside Australia for four hours.

[39:07] That's funny.

[39:09] Oh, that's good.

[39:10] That's good.

[39:11] Well done.

[39:11] Cool, cool.

[39:13] Mason Desmond Lee, any, any updates from the week?

[39:20] Don't hear anything.

[39:23] Joshua Parker, any updates?

[39:26] Anything?

[39:27] Um- Yep ...

[39:27] not much.

[39:27] Just been doing some more homework and tasks I need to do.

[39:32] Oh, okay.

[39:33] Oh, okay.

[39:34] Just busy with your assignments and things?

[39:36] Yeah.

[39:37] Oh, I can see.

[39:38] Okay.

[39:38] I think Mason has responded.

[39:39] All right, Mason, that's fine.

[39:41] As long as things are going good.

[39:43] Okay.

[39:45] I, I know, like, um, we, we next couple of weeks we are, we are...

[39:50] Sorry, we are doing all these sort of things.

[39:53] Um, like, uh, our pretty much P tasks are up until week eight.

[39:59] So we are in week nine, right?

[40:04] Today?

[40:06] Guys?

[40:07] So we are in week nine, and we have week 10, and we have week 11 we have interview.

[40:15] So if you are looking for, basically, uh, can I ask you how many of you are looking for or any of you are looking for HD here?

[40:28] Catherine, you're looking for HD.

[40:30] That's great.

[40:31] Uh, if you are, uh, the rest if you are not looking for HD, that's fine.

[40:35] Catherine, who is your tutor?

[40:37] Is it me or someone else?

[40:44] Sorry, my tutor or the- Yeah ...

[40:48] the marker on- E- E marker.

[40:48] I mean, like who is giving you task?

[40:50] Oh, Zara.

[40:51] Zara.

[40:52] Zara.

[40:53] Okay.

[40:54] That's, that's interesting.

[40:56] Anyway, I think, uh, I'm not sure.

[40:58] I was a bit confused with the...

[41:00] I think, um, Fatima has advised, like, people who are going for HD, they should, of course they should go for interview in week 11.

[41:10] But if Zara is your, is your, um, I don't know, if, if she's your tutor, ideally she should, she should grade you.

[41:19] That's the ideal thing.

[41:20] That's what, that's what I know.

[41:22] But- I just assumed that you would be doing the interview given it's in the seminar time.

[41:27] I think, I think that's what, that's what later Fatima has informed us that in the seminar time.

[41:31] So- Okay ...

[41:32] basically, basically, anyway, if you, if you come to me, if you come to, if you attend in week, uh, week 12, uh, sorry, week 11 you, you can, you can prepare yourself and, and we can go for interview.

[41:44] That's it.

[41:45] Okay?

[41:46] Yep.

[41:46] Yeah, agreed.

[41:46] The rest, the rest all of you are here, if you are not going for HD you don't need to come for interview.

[41:53] That means actually week 11 you guys not, don't need to come here.

[41:57] That's the, that's the simplest thing I can say.

[42:00] And as I have seen up until now, um, uh, we had actually past task up until week eight, so which we have pretty much covered and, and we have done already completed.

[42:12] So maybe I'm a bit ahead while explaining all the weeks' tasks, but, but eventually they are not that difficult.

[42:19] Anyone have any questions up until week eight?

[42:24] Any questions?

[42:26] Because week nine we have actually HD tasks, so, uh, what I will pretty much do, I will try to basically explain, uh, what this HD is, is looking for and some of the, uh, going through some of the descriptions of

[42:39] it.

[42:40] Meanwhile, if you have any questions you can ask me But before I, I start to describe on, on those HD, do you guys have any questions?

[42:49] Please let me know.

[42:50] You can turn on your mic and let me know, or maybe you can raise your hand, we can discuss on, on any questions if you have.

[42:58] Okay?

[42:59] Otherwise, what I will do...

[43:00] Uh, yes, Catherine, go ahead if you have any question.

[43:04] I'd just be keen to get any of your suggestions about how best to prepare, um, for the interview.

[43:10] Like, in the interview, is the interviewer gonna be like, "Okay, I want you to change this code to XYZ," uh, and you have to do it on the spot or?

[43:23] Um, ideal thing is, look, it's not the things that, uh, the, uh, the main thing for this interview is to, first of all, to give you a bit of, a bit of, uh, exercise kind of to prepare yourself for your

[43:42] code, for your demonstration, for your project.

[43:45] This is one main thing.

[43:46] Actually, that is the main purpose.

[43:48] So that in future if you go to some interviews, uh, or some...

[43:53] You attend anywhere, any presentation, you, you are, uh, you have that, have that practice in your mind, "Okay, there will be questions, and I will be able to answer it, and I'm ready for it."

[44:05] Just to give you that practice, that is one of the key part of this HD.

[44:09] The other thing is just to see how confident you are the code that you have written.

[44:14] So probably, for example, you have...

[44:17] You, I can see you have completely working code.

[44:19] Things are working fine with no errors, no- nothing, uh, at all.

[44:25] Things are going fine, and, uh, we get the output.

[44:28] So probably some, some of the things we will, we will be interested to know, what is your understanding of the code you have written?

[44:36] So can you explain me these things, what you have done?

[44:40] Right?

[44:40] You got my point?

[44:42] Yeah.

[44:42] So- So- ...

[44:43] some of it like that will be what we're doing in the video that- Exactly ...

[44:47] we've attached as well.

[44:48] Exactly.

[44:48] Yeah.

[44:48] Okay.

[44:49] Yeah.

[44:49] Exactly.

[44:50] And this is the, this is the one main thing, like, uh, to ensure that you have done your work.

[44:55] Yeah, you can take help from AI, whatever, but you understand it.

[44:59] Do you understand it?

[45:00] Okay, I have used that.

[45:01] Do you understand it?

[45:02] That's the main thing.

[45:04] Yeah.

[45:04] And why have you done it?

[45:06] Can you- Yeah ...

[45:08] if something else, will it be what?

[45:11] How it would be?

[45:12] So we want to gauge a bit of your understanding.

[45:15] That's the main thing.

[45:15] So as I said always, these days AI is doing and we are doing, sometimes we are copying from AI, and we are saying that we have done it, but how far we are confident that this is the one we are

[45:27] looking for.

[45:29] And if sometimes you might be asked, or I might ask you like, "Hey, can I change this one to..."

[45:38] If we change this to what might happen?

[45:40] You just not...

[45:41] You may try like coding that on the fly, or you might just give it, give us an answer of your understanding.

[45:47] That's fine.

[45:49] Uh, in my case, I always ask my questions like in not such that, okay, I know that coding requires maybe 10 minutes or 20 minutes.

[46:02] I don't ask that kind of questions that he might need to.

[46:05] I normally ask the questions maybe you need 30 second to answer.

[46:10] That's it.

[46:11] 30 second to modify.

[46:12] That kind of questions I do normally ask, mostly in case of logical, logical things, like how logically you are, you, you are dealing these sort of things.

[46:21] Not, not just coding level.

[46:23] Coding you may take time, you can use AI, but the logic things is the main thing, right?

[46:28] You got my point.

[46:29] Yeah.

[46:29] Okay, cool.

[46:29] So you can sort of talk through your log- if you sort of said what would you do in this case- Exactly ...

[46:34] or what if I change this, you could talk through the logic of how you would approach it.

[46:38] Yeah, you can.

[46:39] You- Not necessarily have to code it on the spot.

[46:41] Yeah.

[46:42] Yeah, yeah, yeah.

[46:43] Yeah.

[46:43] That's right.

[46:44] Look, um, I don't know you are, you are aware of this sort of things like, um, um, on the spot coding is, uh, is even, uh, Google or big companies when they, they go for interview, actually they will ask you...

[46:59] Have anyone heard about leet code?

[47:02] Anyone have heard about leet code here?

[47:04] Yes, I've heard about leet code interviews.

[47:06] Oh, okay.

[47:06] I think, I think Lachlan has heard.

[47:08] Okay.

[47:08] So, so this kind of some of the coding you will see is a very, very, m- it's too much basics and very fundamentals.

[47:16] So sometimes they give these kind of codings, uh, or, or questions to the, to the in- to the interviewee, um, and, and, uh, the questions are, are really very simple and basics.

[47:29] And sometime you don't need to write the code.

[47:32] They will say, uh, how we you gonna answer it?

[47:34] You need to tell, okay, I'll first do this, I'll first do that, and next I will do this, and then you need to just pretty much write in the bl- uh, whiteboard with the marker, and you tell us the approach.

[47:45] So they basically want to know, they will, they will basically, they basically want to know what is your approach to deal with this kind of scenario, this kind of problems.

[47:54] That's the, that's the key part of it.

[47:56] And sometimes the question might be completely open question.

[48:00] It's an open question actually.

[48:01] Nobody have solved this till now.

[48:03] That's the open questions.

[48:05] Do you have any thoughts on it?

[48:07] What is your thought?

[48:08] That kind of questions are normally asked.

[48:10] Yeah?

[48:11] So Catherine, I think probably you got the pretty, a bit of idea, right?

[48:16] Yeah, no, that's very helpful.

[48:17] Thank you very much for that.

[48:19] No, no problem at all.

[48:20] No problem at all.

[48:21] Yeah.

[48:22] So even, even sometimes I remember I asked some of my students a question, "Can you, can you try, um, you have created a triangle or something with the stars, and there are four or five colors you have applied.

[48:35] Uh, how about like let's change the code and just only one color.

[48:39] Can I do that?"

[48:41] So maybe very, very basic questions I ask them, but, uh, sometimes that take a bit of time.

[48:46] "Hey, how can I do this, and can I do..."

[48:49] Sometimes they go very straightforward way to answer, "Okay, I will put all four, five colors.

[48:54] I used actually five colors here or four colors."

[48:57] And I will just put all the colors into, uh, blue.

[49:01] Let's say they want only blue stars.

[49:03] Then I probably ask them some question like, "No, without changing this piece of code, can we do this?

[49:10] Can we do change the answer?

[49:13] Can we get that answer?"

[49:16] So this kind of like logically, a bit of logical things that we try to give to just to try how, how you think on the spot, and what is your way of thinking.

[49:28] That's the key part of it, actually.

[49:30] Yeah.

[49:30] I think, I think that should be- Thank you.

[49:32] Yeah.

[49:33] Fantastic question, Catherine.

[49:35] Don't worry.

[49:35] Don't take stress.

[49:37] I'm not here to give stress to the people who came to interview, just to test whether actually you are confident on what you have done.

[49:44] That's very important.

[49:46] Okay?

[49:48] Cool.

[49:48] Anyone have any other questions?

[49:50] If not, I will just go through a bit of on the HD one task that we are trying to do.

[49:57] Um, that's, that's what I will probably...

[50:01] Cool?

[50:02] Okay.

[50:03] So that means I will not run any code.

[50:05] I will just open the, the, the Python files, and I will try to explain that to you guys.

[50:11] So let me share my screen, and then I will go, um, uh, and I will go through the, those sort of things.

[50:20] And in between, if you guys have any questions, please do let me know.

[50:24] So I have downloaded, actually, in week nine, we have workshops.

[50:28] I have downloaded, actually, that workshop folder.

[50:31] Uh, in the workshop folder, basically it is, it is asking us mainly to do some kind of things like, um, again, lecture, if you need to-- need the lecture recap or anythings, and H, high distinction task one.

[50:50] So what exactly-- And the other thing is actually here you can see AI and ethics activity.

[50:57] Basically, you can, you can think about like, um, I, I, I believe Fatima has already explained during the lecture.

[51:05] So when you are trying to use AI and make sure, uh, how far, how far you are using AI and, and how does it, does it impact in the, in the socio-technical concept, how far it is fair.

[51:22] Okay.

[51:22] Fairness, okay, when you are trying to do, uh, use AI, how fair?

[51:27] First of all, any problem at what angle you are checking it.

[51:31] Okay?

[51:32] When you are given HD task, high distinction task, or I mean distinction task or any task, how we are thinking the task.

[51:39] First of all, you think about the problem, how can you break down the problem, and how can we do it step by step.

[51:47] Okay?

[51:48] So when you start to think about the problem and you break down the problem, that is one of the key part you can do.

[51:54] Okay?

[51:55] And even breaking down them is also you can do in a way that is more efficient.

[52:01] So these are the parts are very important, and these are the things actually you should use to beat AI.

[52:09] Later, any specific things that AI does, you can cross-check whether AI is doing in a right way.

[52:16] Okay?

[52:17] You can try your own thoughts and thinking, okay?

[52:21] That is one of the key thing you can try and, and see how, how AI is responding.

[52:28] Or sometimes you can see, uh, if AI is actually explaining how it is solving that particular problem.

[52:37] So, so try to use, use in a, in a, in a very, um, how can I say?

[52:43] Uh, in a watchful way, we need to use AI.

[52:46] Um, otherwise, if you leave everything to AI, pre-pretty much we are, we are not using our cognitive skills or cognitive power.

[52:57] Now, I have downloaded the resource file from OnTrack.

[53:01] OnTrack have HD one resource file.

[53:04] And as u- as usual, you can see some of the, some of the things here.

[53:09] And you have been given a presentation guideline, how you will present.

[53:14] Okay?

[53:15] So make sure when you try to work with HD task, uh, HD and D task is pretty similar actually, but when you are doing HD task, um, I think, I think this is a bit, uh, it should be HD one.ipnb,

[53:36] I believe.

[53:38] Because for this is we are for HD, HD task.

[53:43] So that means ideally in the, in the file that is mentioned here, I would say I have to edit that to, that should be, that should be HD one, right?

[54:01] So this is one of the thing we have to keep in mind.

[54:04] Um, you should use Jupyter Notebook as the basic-- basis for your presentation and explain your code and thought process.

[54:11] As you see, you can see thought process is important.

[54:15] Uh, what are the thought process you have applied here?

[54:18] So this is very important.

[54:21] Okay?

[54:22] Thought process.

[54:25] Uh, thought process is extremely important.

[54:27] How you are thinking is this is the ma-ma-mainly key part of it.

[54:31] Recording must only capture you submitted, as I said, this is a bit of correction, uh, we have to keep in mind, on your screen while presenting.

[54:41] You are not required to show your face, which is good.

[54:44] The presentation must not exceed twelve minutes.

[54:46] I think twelve minutes is too much time.

[54:48] If, if I say it should be three to five minutes should be enough.

[54:55] But make sure penalties will be applied for additional time, so I don't recommend you guys to go over twelve minutes.

[55:02] Even, even listening to someone over twelve minutes is too difficult these days, correct?

[55:07] Even if I am talking to you, if I try to explain you something for twelve minutes, it's hard to concentrate.

[55:15] So better it is be precise.

[55:20] Prede-presentation must be recorded using your own voice with clear audio because I, I noticed some of the student try to use, uh, AI, um, specifically international students have done that I have seen before for other unit.

[55:36] So use your own voice.

[55:38] Um, uh, of course, local will not have that issue, I believe, but sometime international think that, okay, I-- my English is a bit poor.

[55:48] Maybe how about I use AI?

[55:50] Don't use AI.

[55:50] It's fine.

[55:51] We are human.

[55:53] Diversity, diversity of the beauty is the beauty of, of human, actually.

[55:58] Uh, take-- accept it, and that's the-- that's, that's awesome.

[56:03] Uh, recording must be uploaded, so you can use any video repository, which is YouTube.

[56:08] You can, you can follow this.

[56:10] Uh, a simple-- If I make it very simple for you is basically if you do the recording, you can use YouTube channel, and you can-- you should not publish it.

[56:20] You can make it unpublished.

[56:22] You can upload the video, but you can make it unpublished.

[56:25] However, you can share that link to anyone.

[56:29] So, so those persons you share with the link, uh, with those person, if you share the link, they will be able to only see that, uh, recording.

[56:41] Presentation must contain following parts exactly in the same order.

[56:45] So the order is very important here as, as, as written here.

[56:50] Uh, you start with the introduction, tell about, about yourself and student number.

[56:55] Okay?

[56:57] Uh, provide an overview of the data science task you are done.

[57:00] So what are the things?

[57:02] You just give a bit of overview.

[57:04] So keep it in, in one minute time.

[57:07] Don't stress too much.

[57:08] Sometimes some student stress too much with the, with the time framing here.

[57:14] Uh, don't stress too much.

[57:15] If it is one minute, three second, five second, we don't mind.

[57:18] Don't to make it two minutes.

[57:20] That's the main thing.

[57:21] Okay?

[57:22] Or make it in between, like, fifty second to one minute.

[57:26] That's it.

[57:27] All right?

[57:29] And then talk about the data cleaning and preparation.

[57:33] Uh, here, actually, you will be talking for two minutes, yeah?

[57:38] But in practical scenario, actually, in any data science project, the cleaning and preparation actually takes at least seventy to eighty percent of the time

[00:00] That's the real-life scenario.

[00:02] But here, when you are presenting, make sure you go around two minutes.

[00:07] That's the thing.

[00:08] Okay?

[00:09] Walk the audience through your code and explain the steps you took to clean, what, what steps you followed to clean and prepare the data for analysis.

[00:20] So b- that means imagine yourse-yourself as a, as, uh, as a data scientist.

[00:25] You are working on a project.

[00:27] You start with your name, student ID, and a bit of your introduction or-- and tell us about, uh, the data science work you, you have done.

[00:36] Uh, kind of your, your own...

[00:38] You are trying to, uh, demonstrate a small project, uh, as a data scientist to the audience, to the stakeholders, something like that.

[00:49] What are the cleaning and preparation you have done for the data?

[00:53] Uh, um, you can share the screen, you can explain the code, uh, share the steps.

[01:00] Don't, don't try to, um, go too much detail of the code level.

[01:08] Too much detail, uh, because, uh, uh, in that case, you'll not be able to cover maybe in two minutes' time.

[01:15] Okay?

[01:15] Uh, go to a bit of high level, uh, your code and explain the code or maybe any specific function you have used or any specific, uh, method you have used or any specific package you have used.

[01:29] You mention that and make it clear.

[01:31] Okay?

[01:34] If this is for part two.

[01:36] Right.

[01:37] And then next step is maybe you talk about your data analysis.

[01:41] This is the core part of it, which is eight minute pretty much.

[01:44] I think eight minute is quite enough, quite enough to discuss the, uh, the, the whole data analysis things.

[01:54] Okay?

[01:55] For each question, walk...

[01:57] So there are something in your, uh, Python file, okay?

[02:02] In H-HD One Python file.

[02:06] So there are some sections of course over there.

[02:10] You can probably e-explain each question and walk through the code and explain it.

[02:15] So explain each of the step, what the code has done, and probably any outcome you received, you explain that.

[02:24] Present the result of your analysis using plots.

[02:27] So one is the code explanation or the process of the code, and the other thing is the plot or table, your presentation.

[02:36] Okay?

[02:37] And finally, your observation.

[02:39] What is the your observation?

[02:42] What you have found?

[02:43] What is the outcome you have found?

[02:45] What observations?

[02:46] Okay?

[02:48] And maybe, maybe give us a bit of creative your observation.

[02:58] Okay?

[02:59] We all are human.

[03:00] We walk through the street or we look around the nature and everywhere, but different people have different observation.

[03:08] Correct?

[03:09] We want to see your observation here.

[03:12] So try to be creative, try to be critical, uh, when you present your observation.

[03:17] That's important.

[03:19] Okay?

[03:20] And this is the things makes you different from, mainly makes you different from others.

[03:24] Where you do the, uh, share your observations, where you show your plots.

[03:30] These two things makes you very different from the other people.

[03:34] Okay?

[03:37] Guys, are you with me?

[03:38] Is that making sense all right?

[03:42] Catherine, I believe sounds all right.

[03:45] Okay, excellent.

[03:47] Yeah.

[03:49] So, uh, why I'm just focusing on these things, there is a big reason behind it, is, um...

[03:57] And also try to show, as I said, try to show a bit of creativity here in these two places.

[04:03] They are important.

[04:04] And observations are important.

[04:06] The reason is like, look, the people in the, uh, in the planet, there are various different people.

[04:12] But based on their observation, based on their, uh, way of the, the think makes them different from others.

[04:19] And that's how actually, uh, Sir Isaac Newton was able to discover, uh, the laws of gravity because he was sitting under an apple tree and then he-- an apple fall from that tree to, uh, to him, and he was thinking,

[04:35] "Why apple is coming down?

[04:37] Why it's not going up?"

[04:40] Okay?

[04:41] Uh, we all know that something actually from top, it falls towards the ground.

[04:46] We know that, but, but his observational, his thinking was different.

[04:50] That's why he has discovered the laws of gravity, right?

[04:54] And on the other hand, Albert Einstein was observing a bit different, right?

[04:59] Different things.

[05:00] That's how he has discovered theory of relativity.

[05:03] And he was observing like how the universe is working, and he was observing that time in the, uh, the, the Earth is not same as time in other, other planet.

[05:14] When you travel with the time of speed, if you go far, far, far from our galaxy to other place, maybe time will be different.

[05:22] Okay?

[05:23] Maybe if you go to a black hole, the time is completely different than the Earth itself.

[05:29] Okay?

[05:30] So that's how the observation makes a human different.

[05:34] So that's why I highly recommend you, uh, during plotting and observations, try to be a bit creative.

[05:43] Challenges and obstacles.

[05:44] Of course, while doing these things, you might have faced some of the challenges, some of the issues, some of the things that seem to me, to you, maybe from the dataset front, from the outcome front, from the coding front, you

[05:59] have received some of the observations or some of the challenges.

[06:04] You can probably explain that, how-- what, uh, challenges you have encountered And how you overcame those challenges.

[06:13] Okay?

[06:14] Possibly you can say also, uh, how things could be better.

[06:21] I think that's a bit of general guideline of how we do submit or how we do present our video.

[06:30] Um, now I will probably go through a bit on the code and try to explain you, uh, a bit on our expectations.

[06:42] So I'll close it up until now.

[06:44] If anyone have any questions, um, even Catherine or anyone, if you have any kind of questions, do feel free to ask me.

[06:53] I'm happy to, happy to, um, explain that to you guys.

[06:59] Okay?

[07:01] Let me know if you have any particular question.

[07:05] So what I will do, I will close this screen now.

[07:08] Uh, close this one.

[07:10] Um, now I will probably go back and see task completion report.

[07:17] I'll try to explain a bit, and then I will go try to explain with these three files.

[07:23] Um, task completion report is pretty much as usual.

[07:30] Whatever you have, um, whatever you have done, and, and the questions are very clear or items are very clear.

[07:40] Uh, make sure you in some places where you need to code attachment, okay?

[07:48] You need to attach the code and a response to the video link because it is hedged data.

[07:55] So you might need to make sure you have done that right.

[07:59] When you-- some of the students have attached the code so far I have seen, but sometimes I can see that I need to request for the, for the access to that code again.

[08:11] So better when you share the code, you give the-- give already, uh, access to your teacher.

[08:20] Uh, if it is OneDrive, you can give access to the teacher, or you can create the file or link in a way that does not require in any permission.

[08:29] So that becomes easier for us to check.

[08:31] I was checking some of the credit task.

[08:33] I noticed that I need to again, um, request for permission.

[08:39] Then a student need to grant the reque-- uh, grant the permission to access the file.

[08:44] Then I could access, and then I need to do it, uh, check it again.

[08:48] So that's cut.

[08:49] Coming and going back and forth takes a bit of time.

[08:53] So this is the, the things.

[08:55] Make sure you do that.

[08:57] Um, again, you write the things in your own words.

[09:01] That is the best thing you can do.

[09:05] One other thing actually I want to highlight here.

[09:11] Um, I'm not sure.

[09:13] Many student take it a bit boring to write the report.

[09:18] Uh, this report writing.

[09:20] Many student I have noticed that they find it very boring or sometimes what they do, they pretty much repeat the previous things and copy-paste, uh, or they take it very lightly.

[09:33] Of course, every task you need to have your, your own observation and individual, uh, way of, of writing the things.

[09:42] Because as you will notice that while writing code, you think one way.

[09:48] While writing response, you think another way.

[09:51] So your brain works differently.

[09:53] Even you will notice when you write in pen and paper with, with the-- with pen and, and a paper, your brain works differently.

[10:02] So see where your words more...

[10:04] Your, your, your thinking are more creative and more, uh, better.

[10:11] You, you try to follow that track.

[10:12] I recommend that one for you guys.

[10:16] The other thing I would say, this is a good skill you should practice from now on.

[10:24] The reason is when you will...

[10:25] I met a-- I met an alumni in, in, in that panel, uh, that I was talking to you, uh, today.

[10:36] Uh, the panel with the students.

[10:41] Uh, so I met an alumni, and she work in a company as a, as a data scientist.

[10:48] So, so she was telling-- I was asking her, actually, uh, she came to see me again at Deakin last week, and she met me, and I was having a good chat with her while in a, in our, uh, while taking

[11:03] coffees, and we are having.

[11:05] I asked her, like, "What, what made you or what skills in the university really helped you when you entered to this job?"

[11:13] So one of the things she was mentioning that writing report actually helps-- helped me a lot in, in some cases when we try to write the code as, uh, we-- our brain works in a very technical way.

[11:28] So basically, the thing is that the idea here is as a, as a human, when we are trying to, uh, for example, I'm a mathematician or, or, or I, I'm reading chemistry, you will feel like these things are working with

[11:46] symbols.

[11:47] Similarly, when you do coding or programming, you feel like, "Hey, this is boring," because things are really weird language, programming language we are using and we are working.

[11:59] But this report is giving you an opportunity to think m- as a, not as a non-technical person and explain the things to a non-technical person So which means you shift your brain or it shift your thinking from technical the- to

[12:17] non-technical or a business perspective.

[12:21] So that's what is actually you develop the skillsets, and when you go to the industry, you eventually need to write a lot of reports on your findings, on your outcomes, on your deliverables, um, on your, on your result to the,

[12:36] uh, stakeholders.

[12:37] Because a stakeholders can be an investor, can be a business owner, can be a, uh, um, another person who has no technical knowledge.

[12:49] So what we want to make sure we are...

[12:51] if we are able to efficiently communicate with those stakeholders who does not have any technical background, and we should be able to explain to them what I have done.

[13:03] Okay?

[13:04] Communicate with them in a written report.

[13:07] So, so these skills are extremely, extremely important.

[13:10] Please don't take it lightly.

[13:12] And when you write the report, you make sure you understand it and, and you are able to explain it, um, to a very general person who probably may not have the IT background.

[13:28] I think that's, uh, that's all about the report.

[13:32] If you guys have any, again, any questions, please let me know, but don't forget to share the video link as well as share the code link.

[13:41] And if you share the code link, make sure the code is accessible.

[13:46] Uh, that's make it accessible at first shot so that anyone, uh, go through your code, they should be able to execute it, or they should be able to run it easily.

[14:01] Uh, any questions up until now?

[14:03] If not, I will try to go to the code that has been shared with us.

[14:09] Anyone have any particular questions up until now?

[14:14] I don't see any questions so far, so let's go back to the code and let's try to see, uh, if we can discuss something on that.

[14:29] Um, so we have-- Have a look at this.

[14:33] We have mainly these three files: HD one, credential-sample.ini, and collect.ipynb, which is the Jupyter Notebook file.

[14:47] So two Jupyter Notebook file and one the credentials.

[14:51] I think let's open those things.

[14:54] Um, and first of all, today I'll, I'll not use Jupyter Notebook.

[15:01] I'll ju-use any online platform to open those files.

[15:05] So here I have that platform ready.

[15:09] I have used, uh, Google Colab to, uh, open those file.

[15:19] So let me, let me share that file.

[15:25] I think you can see now.

[15:26] So this is the Google Colab just to sh- open those files.

[15:29] That's why I don't need to use, uh, Jupyter Notebook for now.

[15:34] Uh, uh, this is eventually a Jupyter Notebook platform.

[15:38] So here, here, here what we are asking, analyzing YouTube video data from news publishers.

[15:47] So there, there are different news publishers, and we are trying to analyze YouTube video data.

[15:53] What is here is this task focuses on analyzing YouTube videos about a news topic, that is COVID-19, published by four English ne- news publisher.

[16:06] So this is a video talking about the COVID-19 in twenty-twenty, actually, and, uh, the news publisher have some video channels.

[16:18] Usually mo- all the news publisher have their own video channel.

[16:23] So, uh, and then, uh, from a start date to the end date.

[16:29] Keep in mind start date to end date is very important to present something.

[16:36] You will need to specify the news topic, so topic I have to specify, and the list of keywords to search for the videos about that topic on YouTube.

[16:47] So basically, every news channel have their-- every news publisher have their news channel in YouTube, which means we can watch the news from YouTube.

[17:00] That's the simple thing.

[17:02] Now we need to focus on at least four publisher and some from some specific date.

[17:13] Okay?

[17:14] And we need to specify the topic and the list of keywords to search for the videos about the topic.

[17:23] For example, the topic is COVID-19, so we need to put some keywords because normally anything, for example, if I want to search anything about AI, we need to put some keyword in the search bar.

[17:40] So based on that keyword search bar, we put the keyword in the search bar, YouTube search bar, and it brings a list of videos from that topic.

[17:53] Correct?

[17:54] So understand first the context.

[17:57] Also, you will need to extract the channel ID.

[18:00] So every channel have their channel ID number.

[18:05] Actually, when we see if it is BBC, Fox News or whatever, we see they have their channel ID, specific ID number Even if you create your own YouTube channel, you will have also a ID number for that channel.

[18:25] Okay?

[18:27] So channel IDs of news publishers using the function `get_channel_info`.

[18:32] So some fix function is already given to us, uh, given to you, attached collect.ipynb notebook file.

[18:43] Okay?

[18:44] So that collect file is here.

[18:47] In this collect file has actually you can see, um, the, the channel information is there.

[18:58] Okay?

[18:59] So, uh, can I see?

[19:05] Get.

[19:09] You can see, uh, `get_channel_info`.

[19:13] You can see that, right?

[19:15] So we have that information is there.

[19:18] You can use `get_channel_info` to extract channel IDs of sample video from news publishers.

[19:26] So you can see that we have been given there, right?

[19:31] Uh, so this is, this is important and this is interesting.

[19:35] So we have been given something here.

[19:40] Now, let me go back and have a look at that.

[19:51] Given to the attached file, all settings need to be specified in section B: Settings of Collect.

[19:59] So we need to go to the settings.collect.

[20:02] So there are actually collect file is there.

[20:04] You can see here.

[20:06] So keep in mind, I-- if I can make you very simple-- if I can make the things very simple for you.

[20:12] HD one is the high distinction one question file, where you are pretty much showing the output, showing the activities in this file.

[20:26] However, collect.ipynb, that Jupyter file is pretty much is the, uh, settings of data collections.

[20:37] Okay?

[20:38] This is pretty much a setting file you can say data collections.

[20:42] All right?

[20:44] So that's what it is.

[20:46] All settings need to be specified in section B: Settings of Collect file, and also copied into, into section B: Settings, uh, of this Jupyter notebook.

[20:59] Okay?

[21:01] And please note that the term news publisher and channel have been used interchangeably.

[21:10] So sometimes we have used news publisher.

[21:13] Basically news publisher, each news publisher have their channel.

[21:17] So this has been used interchangeably.

[21:20] Uh, each news publisher has a channel.

[21:23] If a news publisher has more than one channel on YouTube, pick any.

[21:27] So basically, you can choose any of them.

[21:30] Okay?

[21:30] As stated in your task completion report.

[21:33] The following instructions convey important information and read them.

[21:37] So for example, we have mainly, mainly I would say, we have mainly three files.

[21:44] Uh, we have discussed already, uh, directions and video presentations.

[21:50] The-- but mainly we have that one is the python collect, uh, .jupyter notebook file and credential-samples.ini.

[22:00] So these two are the main file, and another is the hd1, the Jupyter notebook file.

[22:07] Okay?

[22:09] Please watch video about sentiment analysis.

[22:13] So this is about, I don't know, you guys have...

[22:15] Anyone have idea about sentiment analysis?

[22:27] Okay.

[22:30] Anyone here idea about sentiment analysis?

[22:40] Okay, I'll try to explain it here.

[22:43] Uh, you can watch this video.

[22:45] Sentiment analysis is actually nothing but when people express their sentiment in the text, wr- written text format.

[22:54] So how to understand when people post some review comments or post some of the things, um, how people, how people express their sentiments.

[23:11] Means the person is worried, stressed, confused, or, or the person has, has some kind of confusion or angry or happy or satisfied.

[23:24] How can we understand all these sentiments?

[23:28] Okay?

[23:28] This is all about that.

[23:32] Rename that credential.sa- credential-sample.ini file.

[23:36] Basically, this file was you are asked to rename it.

[23:38] The reason behind it, um, the reason behind it is, uh, this file, this credentials.ini, this is, this is the sample file given, but this credential file actually require you to put your information.

[23:58] Uh, let me try to do that credential.

[24:02] Open that file.

[24:05] So basically, this is the information has to be, uh, YouTube API service name, YouTube API version name, and developer key.

[24:16] Means, uh, you are the developer.

[24:19] You should have access.

[24:21] As a developer, you should have access.

[24:23] So you need to create a bit of You need to use the key to access the data.

[24:31] Okay?

[24:31] So that is the key.

[24:33] You need a API key.

[24:34] This is mainly an API key you need to have, okay, to access that YouTube data.

[24:43] So create your YouTube API key, and how to create developer key, you can probably watch the video and that video has explained...

[24:52] Let me try to open it.

[24:57] Redirect notice.

[24:58] Maybe you don't want to visit the page.

[25:02] Looks like...

[25:05] Let me see.

[25:13] I really hate that, uh, the things that is called, uh, uh, ads.

[25:24] So how to create YouTube, um, uh, YouTube API key or how to get actually.

[25:31] Uh, you can probably watch any video, or you can probably search here YouTube API key.

[25:41] And different language and different people have explained at different times.

[25:46] See, these are very short, like only one point four four.

[25:50] See, if that lo- short video doesn't work, you can use any long video.

[25:54] Or even if you feel like the video is, is, is, uh, not very helpful for you, you can even ask AI.

[26:03] AI should also help you to, uh, guide you through a step-by-step how to use that YouTube, um, API.

[26:13] So a- API is nothing but in an authentic way, just, uh, if I make it very simple for you.

[26:19] Imagine, um, I want to open a locker at Aus- Australia Post, uh, next to my house, let's say, uh, in, uh, any branch.

[26:32] So I may, may probably need to first put some credential and give my credential to them and, and validate that this is the right person, I am the right person.

[26:41] And then probably based on my all my credentials, they check and then things, and then they are happy to give me a key.

[26:46] "Here, that is your key.

[26:48] You can access."

[26:50] So that's how they make the thing secure and, and you are able to use that space.

[26:56] The same thing here for YouTube for these, uh, places.

[27:01] Uh, they give you access to different videos through this key.

[27:06] That means they understand that this is the person trying to access this data, and they allow you to access this data through an API key and make, make sure that you are not a robot, you are a human who is

[27:20] actually trying to use, uh, that API.

[27:26] Okay?

[27:27] See the following.

[27:28] Okay.

[27:28] And, um, you can view following link for more details, so [developers.google.youtube](https://developers.google.com/youtube).

[27:36] So Google has actually has shared, uh...

[27:39] Let me see the video.

[27:43] So it has shared actually the page, and the page is very clear and step-by-step.

[27:47] Very, very...

[27:48] I would say it's a very simple and very straightforward way, but that gives you quite a bit of detail on how you can use the, use the API.

[27:58] Okay?

[27:59] Resources and resource type.

[28:01] Or resource type means what types of field you can access.

[28:05] You can see in the channel, you can access channel banner, channel section.

[28:10] And basically, after collecting, collecting this data, after collecting this data, you can probably analyze a lot of things.

[28:20] Okay?

[28:21] Even you can analyze, uh, how can I say?

[28:25] A lot of, lot of, uh, forensic analysis also you can do.

[28:31] Okay?

[28:31] So these are the few things.

[28:33] And you can see here, uh, s-supported operation.

[28:39] Look here, they support some operation for general user.

[28:44] Okay?

[28:46] And some operation they don't know- they do not support.

[28:52] Okay?

[28:53] So some operation they support, some operation they don't support.

[28:58] So basically you can see watermark.

[29:02] So s- so some operations are free, maybe some operations if you want to do, you may need to get for subscription or something else.

[29:10] All right?

[29:11] Or maybe take a special permission from them.

[29:13] But ultimately, they have even much more data than we are able to access, uh, about the YouTube channels.

[29:20] Okay?

[29:21] Much, much more data.

[29:22] Even which IP, uh, the person has logged in, what time, all detailed information you can get.

[29:31] Anyway, uh, the thing is, uh, a lot of things are explained here and examples are also given.

[29:37] Actually, this example will help you to understand the things much better.

[29:42] I recommend you guys have a look at these examples.

[29:46] Example one, example two, example three, and example four.

[29:51] A lot of example with the description is also given.

[29:55] Okay?

[29:56] And how the data is being collected.

[29:58] Okay?

[29:59] Specifically, the statistics you can see, view like, view count, and even, even, uh, comment count.

[30:07] Okay?

[30:08] So they have shown a lot of things here and how these things work.

[30:12] You can go through this.

[30:14] I, I believe this is a great resource for your assignment.

[30:19] This is the page.

[30:20] And, and probably you can, you can do a bit, bit more and you can have a look, have a look.

[30:25] These are very well-established resources and very good one.

[30:30] Okay?

[30:31] Uh Uh, authorized res- request, overview, installed apps.

[30:38] Uh, a lot of, a lot of things are there.

[30:41] You can, you can probably go through from data API, YouTube data API.

[30:46] Okay?

[30:46] YouTube data API and a lot of explanations are there.

[30:50] Okay?

[30:50] Um, what else?

[30:53] Actually, some of the big, big people, some of the big, um, how can I say?

[30:59] The, the, the people with more and more subscribers, some of the channels they have more than millions.

[31:07] Some of the channels that has more than millions of subscribers, they may use these APIs and create a clean dashboard, and they can actually have a look at the comments people has given, what people want from them, what else.

[31:23] A lot of information they can collect.

[31:26] Every individual user, okay, can collect this data.

[31:30] So I highly recommend all of you, if you are doing HD1, uh, go through this API and you will get to know a lot of things.

[31:43] Okay.

[31:44] What else I was going to discuss?

[31:46] Okay.

[31:48] So this is the link, is very important.

[31:50] Please note projects that enable YouTube data API have a depau...

[31:56] Note projects that enable the YouTube data API have a default quota allocation for ten thousand units per day.

[32:05] To make an efficient use of your quota, please avoid multiple executions of this.

[32:12] Okay?

[32:13] Which means, which means, um, I can have a look at this one.

[32:22] Have a look at this.

[32:24] To work with YouTube APIs and queries, you need Google, install Google API Python client using this conda.

[32:31] So this is a very clean information given to you how you can install Google API Python client.

[32:39] The reason behind this is you are actually, you as a client, you are trying to use Python code to access to the Google server, which means YouTube, and then try to collect the data.

[32:54] So the idea here is the, the user, when you are sending requests to the, to the API or sending requests to the, uh, to the Go- uh, YouTube API, so basically you need...

[33:09] There is a limit how much s- uh, you need or how much a request you can send.

[33:15] I think I can have a look at this.

[33:17] Uh, so make sure you don't send too much request per day.

[33:24] You can see here, uh, ten thousand units per day combined with, uh, a- all other endpoints.

[33:32] So basically, when you are trying to send request to the, uh, to the YouTube, uh, when you are trying to collect the data from YouTube, they have given you a limit for certain unit.

[33:49] If you try to go, uh, above that, and if you try to send more request, they might say this is...

[33:55] Or there is a throttle effect, or this is probably might be done by a bot or AI.

[34:01] They might completely cancel your API, or they might completely stop you or block you.

[34:07] So again, you may need to open another API and, and, and a lot of things.

[34:11] Uh, sorry.

[34:13] Mm.

[34:13] So this is, this is what we have to really, uh, careful about.

[34:21] So make sure you understand, uh, that.

[34:25] Okay?

[34:26] To work with YouTube send queries you need to use, okay, I have discussed it, and specify news topic and news publishers.

[34:33] Okay?

[34:34] Topic is maybe, for example, COVID-19 or anything you like to, and a list of keyword to find the videos about that topic.

[34:43] So if you are talking about the AI and robotics, or maybe AI and machine learning, or AI and data science, y- uh, you can probably put some, uh, first of all, consider a topic and then put some keywords around that

[34:59] topic.

[34:59] You can extract the keywords by looking at some sample video titles.

[35:05] So keywords, maybe as I said, you can, you can go to maybe, let's say...

[35:10] Oh, sorry.

[35:10] I think I...

[35:11] My battery is a bit down.

[35:13] I have to move and connect my...

[35:16] Can you allo- allow me just a few minutes?

[35:19] Uh, let me bring my charger.

[35:21] Just give me a second.

[36:43] All right.

[37:20] Again, um, okay.

[37:24] I think, um...

[37:31] So I'm discussing here.

[37:33] So again, important is using keyword.

[37:38] Choose at least four popular, we have discussed that.

[37:42] Extract manually the video IDs from sample videos from different publishers.

[37:49] Okay?

[37:50] So you can, uh, you can do by clicking and share button for each video.

[37:58] So probably, um...

[38:13] So, so probably you can try checking it from here.

[38:17] When you try to see the things, you can probably see maybe the channels.

[38:24] Okay?

[38:24] Maybe you can see the channel, how you can try to get the channel information from when you have the share options.

[38:31] Okay?

[38:32] So there are many, many ways you can get the channel information.

[38:36] Um, this is one of the things that is saying, um, you can, uh, uh, manually can extract the channel information.

[38:47] Okay?

[38:48] Extract the channel IDs of the video.

[38:51] So channel IDs, um, we can get the channel ID by, as I said, we can get the channel ID information.

[39:00] So from here, for example, uh, the keywords is given here, the coronavirus, COVID, COVID-19, and pandemic.

[39:09] So you can see, uh, the channels information for Fox News.

[39:15] Uh, let me go to Fox News now.

[39:18] Let me see Fox News.

[39:24] We can try Fox News.

[39:29] Um, okay, let me try to...

[39:38] Hmm.

[39:54] Uh...

[39:55] Okay.

[39:57] Okay.

[39:57] I think Fox News has too many channels, so make sure whatever Fox News you are trying to...

[40:03] Uh, you can see Fox News clips and business there are a number of, but see Fox News.

[40:09] Okay.

[40:10] See like which one you are, you are trying to use.

[40:13] Of course, you can use this channel information from here.

[40:17] Okay?

[40:19] Uh, channel ID, uh, these are the things given to us.

[40:23] All right?

[40:24] Um, maximum videos actually here, maximum videos are fifty.

[40:31] Um, that means the maximum number of video that should be returned for each request.

[40:35] So when you are trying to send the request, maximum fifty videos you can.

[40:39] So that's, that's how you can collect the, uh, uh, the videos for each of these channels.

[40:55] Uh, okay.

[40:56] What else?

[40:57] We are in, um, channel videos.

[41:03] You can extract the chosen keywords and this and channel information.

[41:08] Okay.

[41:09] Collect the video and comments.

[41:12] So these are the two things, videos and comments using collect dot this.

[41:15] So you try to use.

[41:17] So this is, as I said, this is the place which is for mainly the data collections, and this is the place for, uh, for presenting your data.

[41:29] That means you need to do cleaning, you need to do, uh, plotting, and you need to, uh, send your observations, all these sort of things.

[41:38] Okay?

[41:39] Now where we are?

[41:42] Here.

[41:43] So...

[41:51] So videos, collect the videos and comments.

[41:53] So we are asked to use the videos and the comments.

[41:57] Specify channel ID, IDs and names in settings section.

[42:02] So when you try to do the settings, uh, attached in collect dot this channel.

[42:07] So you, in the settings section, we need to use that.

[42:10] Whatever in your case, you can use the, uh, channel ID and channel name.

[42:15] So this is the ID, and that is the name of the channel.

[42:20] Uh, that's what we are asked to do.

[42:23] Uh, whatever data you are trying to collect.

[42:26] So you have been given that collect sample.

[42:28] So data collection, that sample is given to us.

[42:32] Uh, and specify the keywords, what keyword you are looking for.

[42:36] If you, as I said, if you are looking for AI, you need to put the specific keywords.

[42:41] Okay?

[42:42] That's it.

[42:45] Uh, okay.

[42:46] Uh, my suggestion is don't, don't, don't use this default one, the default coronavirus, COVID or pandemic.

[42:54] Use something new, some different one, new one.

[43:00] And a specific start date and end date, because from which date to which date you want that videos from, and specify the maximum.

[43:08] You can extract at least twenty videos per news, uh, channel.

[43:11] So basically, for example, here, one, two, three, four.

[43:16] Four channels are there.

[43:18] Uh, I don't know how many channels you are asked to use here.

[43:24] Um, at least four, I guess, right?

[43:30] Mm.

[43:33] Four channels.

[43:34] So from four channels, uh, this is saying at least twenty-five videos per news publisher.

[43:40] So you can use twenty videos, twenty or twenty-five videos so that, uh, you can see here, it is saying actually fifty videos.

[43:48] So you can use twenty, twenty-five videos.

[43:51] So twenty-five videos, it will be a hundred videos.

[43:55] All right?

[43:56] And what else?

[44:00] What else we are given, I think.

[44:04] And here, the other thing is, make sure for your specified topic, three hundred comments per news channel can be collected.

[44:16] Okay?

[44:17] So whatever topic you are choosing it for your topic, so per news publishers.

[44:25] Okay, you can try to collect.

[44:27] So people are sending comments, right?

[44:30] Uh, so that comments is, um, uh, that comment is basically for what?

[44:37] That comment is for, uh, this comment, the comment that you want to collect from the publisher.

[44:46] Please do not make any changes to the color file.

[44:48] Execute this file to collect the videos will be stored in.

[44:52] So basically, you want to store that video.

[44:55] So the thing that is-- That means ideally here, you are trying to collect the comment, right?

[45:05] You are trying to collect comment.

[45:09] You are trying to collect the videos.

[45:11] So two things mainly you are trying to do.

[45:13] You are trying to collect the videos file, and you are trying to videos.csv, and you are trying to collect the comments.csv.

[45:21] Two files you are trying to collect from four news channels based on the keyword that you give, and you need to put a start date and time, a start date and a start end date.

[45:33] So here, twenty, twenty, twenty twenty and twenty twenty-three, pretty much three years of data.

[45:38] Uh, among these three years, like, try to collect fifty videos, so you can try to collect in your own way.

[45:45] I think that's a good thing we can talk about, and I think that's it.

[45:51] Complete the task.

[45:52] Choose only one question from section E.

[45:55] So basically, you start to work with settings in section B.

[46:00] You put the settings here.

[46:02] How-- what is the keyword?

[46:03] What is the start date, end date, and all information, you put it here.

[46:08] And, uh, for importing, you need a lot of this, um, a lot of these libraries.

[46:17] As I said, these are libraries for natural language processing.

[46:21] So this is called NLTK, means Natural Language Toolkit.

[46:25] And this is for data cleaning, stop words, stop words.

[46:29] Word cloud is used for data presentation.

[46:31] So when you try to present your comments, user comments and everything, and the keywords, probably you can use word cloud.

[46:40] Okay?

[46:42] Uh, calendar, you may need calendar also.

[46:46] For the date formatter, you may need to format the date, and you may need to probably, uh, do a lot of data or tokenizations.

[46:56] So that's why we use NLTK dot stem.

[46:59] From that, we used word lemmatizer.

[47:04] Lemmatizer means word, word lemma.

[47:06] We try to, try to break down the keyword.

[47:10] From the comments, we want to get the specific words from the things.

[47:14] So this is a bit of example given here.

[47:17] Um, and, uh, stop words means actually when we say, for example, write Python code that cleans the data for analyzing the video and comments.

[47:32] So here, basically, the, for that, these are not important wo- important word that carry any meaning.

[47:41] So basically, stop words are the words that actually does not require or does not carry any meaning, so we have to get rid of them.

[47:49] So this is the only idea.

[47:50] So basically, you may need to clean and prepare the comment file.

[47:56] Okay?

[47:57] That's what it is, and that's what it is written here.

[48:01] I think, let me see if there is anything mentioned.

[48:03] Okay, let me first read this.

[48:06] Question E.

[48:07] Okay, we'll come about that.

[48:08] Please note that each question you must briefly discuss your observation.

[48:13] So you have...

[48:14] You are doing cleaning, you are doing D1, D2, and D3, you are doing something.

[48:21] So you briefly discuss your observation, uh, on each of the task.

[48:26] For E, analyze video and comments.

[48:28] Please make sure you have watched this video.

[48:32] Understood sentiment analysis, which means actually, uh, as I mentioned earlier, the sentiment is something what people has expressed.

[48:41] After watching the video, people has expressed something.

[48:43] Oh, they liked it.

[48:45] They hate it.

[48:45] This is very bad This went really well.

[48:49] Oh, awesome.

[48:50] Something people has mentioned and expressed, and sometimes people don't express anything.

[48:54] Oh, yeah.

[48:56] People may say, uh, this, this was maybe no positive or negative, they just describe something.

[49:03] Uh, yes, I found, I found something relevant to my life and blah, blah, blah.

[49:07] They mentioned something, for example.

[49:09] So sometimes the positive sentiment we get, sometime neutral sentiment, sometime negative sentiment.

[49:14] So the idea here is choose only one of the following questions.

[49:20] So you are-- There are a couple of question given here.

[49:23] One of them you may need to, you may need to implement.

[49:29] Only one of the following question and write Python code to answer it.

[49:32] Please note that you must briefly discuss.

[49:34] Again, each time you need to briefly discuss, you need to, uh, some function has been already mentioned here to get sentiment and to get sentiment score.

[49:46] Sentiment means actually, first of all, you get the positive and negative and neutral, but a score means out of, out of one, how much it is positive.

[49:56] Let's say if it is positive is point six and negative is point four and neutral is zero, which means people have talked sixty percent positive in that comment, forty percent negative in that particular comment.

[50:15] So that's what is called sentiment, uh, the polarity of the sentiment, and you get the score of this.

[50:24] Okay?

[50:25] That's what it is.

[50:26] So basically E one to E three, you are advised to respond to one of these.

[50:34] That's it.

[50:35] Uh, so let me go ahead a bit.

[50:43] Briefly discuss observations.

[50:44] Please note that there is no unique solution for most of-- most questions.

[50:49] You-- If you need to make any assumptions, state your assumptions.

[50:54] So basically, uh, there is no specific, as I said, it's, uh, pretty much open to you.

[50:59] There is no specific, uh, solution.

[51:01] If you need to make some assumption, you make it and you go ahead, you complete the project and let us know, I assume these, these, these things.

[51:09] Okay?

[51:11] And, and in writing the report.

[51:14] Please note that if something has not been clearly stated in a question, that is the type of visualization.

[51:20] For example, you are asked to plot it or table or maybe some of the word cloud or something is already, uh, some of the libraries are there.

[51:29] It means you are allowed to make your own choice.

[51:31] As I said, already, uh, the plotting word cloud does not fall under.

[51:36] There was somewhere I was showing the word cloud.

[51:39] Somewhere here, for example, word cloud.

[51:42] So word cloud does not actually fall under any kind of table or plotting.

[51:46] It's a visualization type is word cloud.

[51:49] I want to show, uh, what people talk about, about, uh, this channel.

[51:56] I think I have one printed.

[52:01] See if I have it in my paper.

[52:03] I'll try to show you.

[52:05] Uh, I don't know if I have it or not.

[52:07] I, I was the other day, I prepared a presentation where I tried to create a word cloud.

[52:16] Not sure if I have it.

[52:21] Um, no, I think...

[52:28] Let me see in any way.

[52:30] Okay.

[52:31] I don't know you guys can see it or not.

[52:33] So I have done a bit of analysis.

[52:36] For example, I have been teaching.

[52:39] I don't know you can see or not.

[52:41] I'm not sure.

[52:42] I'll try to just show that in s- in the camera.

[52:46] Let me get out of these things.

[52:51] I don't know if you can.

[52:52] I think probably should be all right.

[52:55] So I was trying to show that...

[52:57] Okay.

[52:58] Um, I was teaching for many years.

[53:02] I have been teaching at university, so I have collected a student feedback about me, and I try to create a word cloud, uh, for myself for last, I think, how many years?

[53:14] Uh, twenty-two to twenty-four.

[53:16] Three years, actually.

[53:18] So I have seen three years of student feedback and collected a student has given me all the feedbacks, and then I created a word cloud.

[53:26] So I, as I can-- as I think probably you can see, I don't know you can see it well or not.

[53:31] So in this, uh, word cloud, actually, um, I try to highlight what are the things a student liked or a student has talked more about me.

[53:42] So one of the keyword I can read, I think three or four keyword that I can see is one of the main c-keyword I can see is the helpful.

[53:53] This is the biggest one I can see.

[53:55] And another is good.

[53:59] So maybe he's good at explaining things, good with example, something like that a student has mentioned.

[54:05] And some other few words, easy working.

[54:08] So maybe explain the things easily or something like some kind of word cloud.

[54:12] Yes, Catherine, go ahead.

[54:16] I just wanted to check for the interview, will we, will we be asked about the code that's already been provided for us?

[54:23] Like the code that creates the, the functions for the, um, sentiment analysis and the cleaning and preparation and that sort of thing?

[54:32] Or just the code that we're writing?

[54:35] Uh, look, these are the things that are given to you.

[54:38] These are the code for collecting the data, right?

[54:42] Uh, some of the codes are for data collections and things are given to you.

[54:46] Okay?

[54:47] And some of the answers you have already created, right?

[54:52] Some of the questions you will be answering.

[54:54] Correct?

[54:56] Yeah.

[54:56] I'm sort of...

[54:56] I'm asking 'cause we, um, can add data into, say, that D1 there if we do additional cleaning and add- Yeah ...

[55:04] data to D2, but there's already...

[55:07] Uh, sorry, add code into those if we wanna do more cleaning and preparation.

[55:12] But- Yeah ...

[55:12] will we be asked questions about the code that's already been provided for us?

[55:17] Yeah, I g- I got the question.

[55:18] So look, the c- code that has been provided to you, have you used that code for data generations or anything?

[55:28] Have you provi- have you used them?

[55:30] If you have used them, right?

[55:33] Or that has- Yeah ...

[55:33] generated something for you, you should ideally understand that.

[55:37] Yeah.

[55:38] Cool.

[55:38] You got the point?

[55:40] Understood.

[55:40] Yeah, yeah.

[55:40] No, I understood.

[55:40] Because if you don't understand how you are able to make some changes over here.

[55:45] Yeah?

[55:45] Yeah.

[55:45] So that's the things, at least, uh, to some level we, we, we would expect that you understood that code.

[55:52] Otherwise, uh, that doesn't make sense, like how you build something on top of that.

[55:58] Got it.

[55:58] You got it?

[55:58] You got it?

[55:59] Thank you.

[55:59] Yeah.

[56:00] Yeah, yeah.

[56:00] Thanks.

[56:01] I think that's a good questions, but I, I can see pretty much many things are given here, and the explanation or the, the things are very well, how can I say?

[56:12] Very clearly mentioned what has to, has been done, uh, what h- what should be done.

[56:16] For example, if something has not been done or stated clearly, uh, type of visualizations, it means you are allowed to make your own choice.

[56:25] So things are pretty much given, read the guidelines, how should be submitted, all sorts of things.

[56:30] And many of the code actually, or I can say hints for the codes are already given.

[56:36] So you just need to know a bit like, uh, not all the libraries, but my suggestion is at least in the code level, for example, video DF, published at work, what it is doing.

[56:48] So I think from the, even from the comment itself, we understand what it is doing.

[56:52] So when you get it to work, we ideally expect that, oh, you understand this, what has been given to you.

[56:59] Yeah?

[57:00] Thank you.

[57:01] Cool.

[57:02] No problem.

[57:02] Thank you so much.

[57:03] I think we are almost there, and we...

[57:06] I will, if needed, I will probably explain again next week few more things, or next week, again, I will, uh, explain, uh, more of these files if you need.

[57:16] But I hope you'll be able to work, uh, already on, on these things.

[57:21] Um, let me know if you have any further comments or any questions we can discuss next week.

[57:26] Okay?

[57:27] That is the last chance we get before we go for the interview.

[57:31] All right?

[57:32] All the best for you guys.

[57:34] Take care.

[57:35] Maybe I will see you next week.

[57:39] Thank you.

[57:40] Have a good evening.

[57:41] See ya.

[57:41] Have a good evening.

[57:42] Bye.

[57:42] Take care.

[57:42] Take care.

[57:42] Bye-bye.

[57:43] Bye.